

KORADA TANVI

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PROFESSIONAL SUMMARY

B.Tech CSE (Cloud Computing) student at SRMIST Chennai, CGPA 9.27/10, building and shipping full-stack AI-native products across computer vision, regulatory compliance automation, and autonomous agents, using React, Next.js, Node.js, and LLM APIs (OpenAI, Groq, Claude, Gemini). Published first-author research on LLM pipeline architecture. Experienced in REST API development, real-time computer vision pipelines, event-driven automation, and cloud deployment (AWS, Vercel, Render, Railway).

EDUCATION

SRMIST Chennai — B.Tech, CSE (Cloud Computing)

2024 – 2028

CGPA: 9.27 / 10

Tirumala Junior College — Class XII (MPC), BIEAP

2024

Score: 95.2%

Ravindra Bharathi School — Class X, BSEAP

Score: 88%

TECHNICAL SKILLS

AI & GenAI: OpenAI API, Groq API, Claude API, Gemini API, LangChain, LangGraph, MediaPipe, YOLO, Computer Vision, Prompt Engineering

Languages: Python, JavaScript, SQL, C, Java (basics)

Frontend: React, Next.js, Tailwind CSS, HTML/CSS, Three.js (React Three Fiber)

Backend: Node.js, Next.js API Routes, REST APIs

Databases: PostgreSQL, MySQL, Firebase, Supabase, Upstash Redis

Cloud & Tools: AWS, Vercel, Render, Railway, Docker, Git, GitHub, Postman, Linux

TECHNICAL PROJECTS

AppCompiler — AI App Schema Compiler

appcompiler-ten.vercel.app

- 4-stage LLM pipeline (Intent Extraction, System Design, Schema Generation, Validation + Repair) converting natural language into complete DB, API, UI, and Auth schemas.
- Built a targeted repair engine that fixes inconsistent schema layers without a full retry; validated cross-layer consistency between API, DB, UI, and Auth.
- Published first-author research paper on the system's architecture and evaluation (Zenodo preprint, DOI 10.5281/zenodo.20644045); evaluated on 20 test cases with 85–90% success rate. Submitted to Microsoft × Jio AIDE 2026 Conference.

SatyaLabel — AI Compliance Checker

satyalabel.vercel.app

- Built for Smart India Hackathon 2026 (Problem Statement SIH26034 — Legal Metrology, Packaged Commodities, Rules 2011) for the Ministry of Consumer Affairs; scans packaged product labels via OCR and validates mandatory declarations against the rules, generating compliance reports with rule citations.
- Built as an installable, mobile-first PWA on a zero-cost stack — Tesseract OCR with Gemini Vision as fallback, Next.js, PostgreSQL, deployed on Vercel + Render.
- Part of a 6-member team that placed 8th among 50 teams in the college's internal SIH selection round; application now submitted to the official Government of India SIH portal for the national round.

ShrimpCount — Automated Hatchery Population Estimation

[In Development]

- Building a YOLO-based computer vision system to automatically count and estimate shrimp population density from hatchery tank images, replacing manual visual counting.
- Designing the data pipeline for model training on real hatchery image data, targeting deployment for MAS Aqua Techniks, a commercial shrimp hatchery (freelance client, LOR issued).

Physio — Real-Time Physiotherapy Coach

physio-by-tanvi.vercel.app

- Real-time exercise form correction via phone camera using MediaPipe pose estimation and joint-angle calculations from clinical physiotherapy literature.
- Detects compensatory movement patterns and rep-by-rep form degradation; fully browser-based PWA with Groq-generated coaching feedback, no download required.
- Added interactive 3D exercise demonstrations using react-three-fiber, mapped to real-time pose landmarks for visual form guidance alongside live camera feedback.

StudyForge — Adaptive Learning System Generator

[In Development]

- Extension of PrepOS: takes a student's study goals and topic list as input and generates a full personalized learning system — structured modules, resources, and assessments — tailored to that requirement.
- Applies adaptive content-generation logic to turn an unstructured study requirement into a structured curriculum, rather than delivering static, pre-built prep content.

Warden — Autonomous Personal Agent

[Live]

- Headless, event-driven personal automation agent using LLM tool-calling across Gmail, GitHub, and Drive; installable as a PWA on both mobile and desktop.
- Job-application tracking module (evolved from TrackR) stable in daily use — automated application tracking, deadline alerts, and pipeline management; agent and integration layer under active iteration.

PROFESSIONAL EXPERIENCE

Software Development Intern | ATRIBS Software Systems Pvt Ltd

Sep 2026 – Oct 2026 (Hybrid)

- Working on Progressive Web App (PWA) development — building installable, mobile-first web applications with offline capability and app-like UX in a live production environment.

Prompt Engineering Intern | Future Interns

Dec 2025 – Jan 2026

- Designed multi-step AI workflows and agent behaviors for GenAI systems across 5+ business use cases using structured prompt templates and contextual injection.
- Optimized LLM output quality through task decomposition and iterative refinement, improving response consistency.

Full Stack Web Development Intern | Prodigy InfoTech

Jun 2025 – Jul 2025

- Built responsive full-stack web modules using HTML, CSS, JavaScript, and REST API integration across 3+ production features with mobile-first UI across 10+ screens.

PUBLICATIONS

AppCompiler: A Multi-Stage LLM Pipeline for Generating Validated Application Schemas from Natural Language. Korada Tanvi. Zenodo preprint, 2026. DOI: 10.5281/zenodo.20644045. Submitted to Microsoft × Jio AIDE 2026 Conference.

ACHIEVEMENTS & CERTIFICATIONS

- Placed 8th among 50 teams in SRMIST's internal Smart India Hackathon selection round (SatyaLabel, SIH26034); advancing to the official Government of India SIH portal round.
- OpenAI × Outskill AI Builders Hackathon — Final Round (shipped StudentOS, a 19-panel AI academic OS, solo in a 7-day sprint)
- Python Programming — DataCamp
- AWS Cloud Fundamentals — DataCamp
- MongoDB Overview — MongoDB (official)
- Prompt Engineering — Future Interns | Web Development — Prodigy InfoTech
- Freelance: MAS Aqua Techniks — delivered marketing website + attendance management tool (LOR issued)
- Winner: Photography Competition — SRMIST ('Saving Nature' inter-department)

CLUBS & ROLES

- Andropedia — Media Team Member
- SlugNPlug — Previous Member (Hardware & Systems Club)
- Camogenics — Photographer, Film Society